

Lakshya JEE 2027

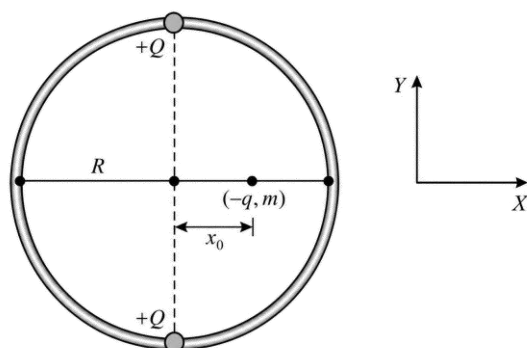
KPP - 04

Electrostatics

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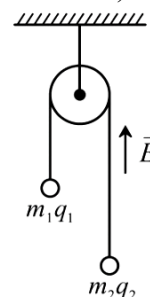
1. A thin insulating wire is stretched along the diameter of an insulated circular hoop of radius R . A small bead of mass m and charge $-q$ is threaded onto the wire. Two small identical charges are tied to the hoop at points opposite to each other, so that the diameter passing through them is perpendicular to the thread as shown in figure. The bead is released at a point which is located at a distance x_0 from the centre of the hoop. Assume that $x_0 \ll R$.

- What is the resultant force acting on the charged bead?
- Describe the motion of the bead after it is released
- Use the assumption that $\frac{x}{R} \ll 1$ to obtain an approximate equation of motion, and find the displacement and velocity of the bead as functions of time
- When will the velocity of the bead will become zero for the first time?

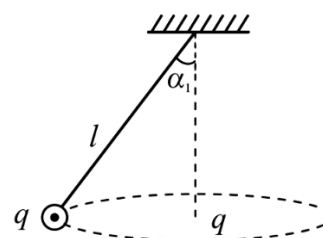


2. A system consists of a thin charged wire ring of radius R and a very long uniformly charged thread oriented along the axis of the ring, with one of its ends coinciding with the centre of the ring. The total charge of the ring is equal to q . The charge of the thread (per unit length) is equal to λ . Find the interaction force between the ring and the thread.

3. Two small balls carry masses and charges m_1, q_1 and m_2, q_2 respectively. The entire system is placed in uniform electric field $\vec{E}$ directed upwards. If $(m_1 + m_2)g = (q_1 + q_2)E$, then the acceleration of centre of mass (Neglect the interaction between balls)



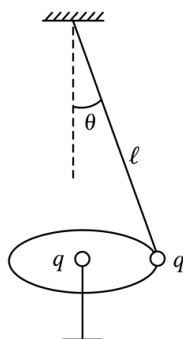
- is necessarily zero
 - is not necessarily zero
 - is directed downwards
 - none of these
4. A charge q is revolving around another charge q as shown in a conical pendulum. The motion is in a horizontal plane.



- Tension in the string is greater than the weight of the ball.
- the tension in the string is greater than the electrostatic repulsive force
- If the charge is removed, the speed of ball has to be increased to maintain the angle.
- If the charge is removed, the speed of ball has to be decreased to maintain the angle.

5. A spherical bob of mass m and charge q suspended from a string of length l rotates about a fixed charge identical to that of the bob (Fig.). The angle between the string and the vertical is θ . Find the angular velocity of uniform rotation of

the bob. It is given that $\frac{Kq^2}{ml^2} = g$.



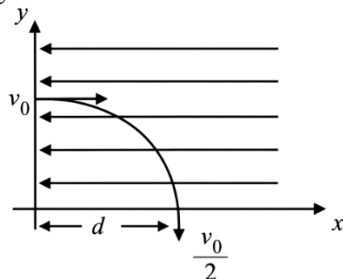
(1) $\sqrt{\frac{g}{l}(\sec\theta - \csc^3\theta)}$

(2) $\sqrt{\frac{g}{l}\csc\theta}$

(3) $\sqrt{\frac{g}{l}\sec\theta}$

(4) $\sqrt{\frac{g}{l}}$

6. In a certain region, free from gravity, electric field is along negative x -direction and it is constant. A particle having mass ' m ' and charge q is projected along x -direction with speed v_0 . An additional force $\vec{F} = \vec{C} \times \vec{v}$ is acting on the charge where $\vec{v}$ is velocity vector and $\vec{C}$ is a constant vector. The charge comes out of region with speed $\frac{v_0}{2}$. Then the magnitude of electric field will be



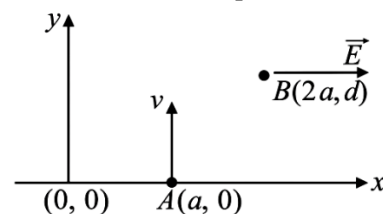
(1) $\frac{3}{4} \frac{mv_0^2}{qd}$

(2) $\frac{4}{3} \frac{mv_0^2}{qd}$

(3) $\frac{3}{8} \frac{mv_0^2}{qd}$

(4) cannot be determined

7. A uniform electric field of strength E exists in a region. An electron (charge $-e$, mass m) enters a point A with velocity v . It moves through the electric field and exits at point B. Then



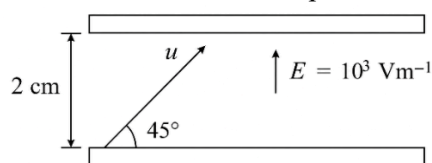
(1) $\vec{E} = -\frac{2amv^2}{ed^2} \hat{i}$

(2) Rate of work done by the electric field at B is $\frac{4ma^2v^3}{d^3}$

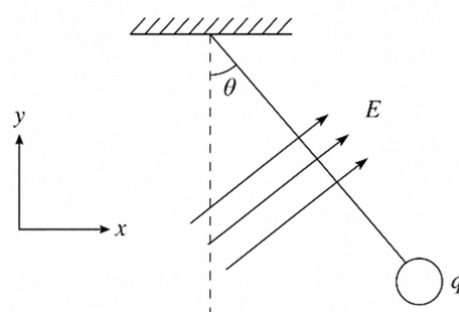
(3) Rate of work by the electric field at A is zero

(4) Velocity at B is $\frac{2av}{d} \hat{i} + v \hat{j}$

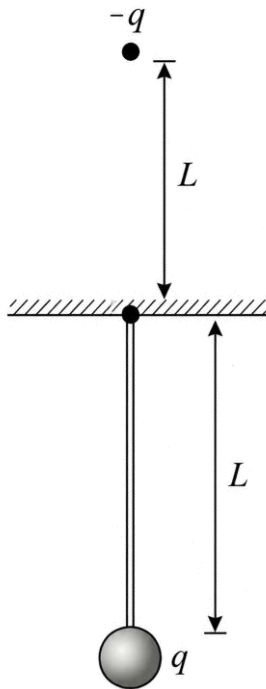
8. A particle having charge that of an electron and mass 1.6×10^{-30} kg is projected with an initial speed u at an angle 45° to the horizontal from the lower plate of a parallel-plate capacitor as shown in figure. The plates are sufficiently loni and have a separation of 2 cm. Find the maximum value of the velocity of the particle so that it does not hit the upper plate Take the electric field between the plates as 10^3 Vm^{-1} directed upward.



9. A charged cork ball of mass m is suspended on a light string in the presence of a uniform electric field as shown in figure. When $E = (A\hat{i} + B\hat{j}) \text{ NC}^{-1}$, where A and B are positive numbers, the ball is in equilibrium at the angle θ . Find the charge on the ball.

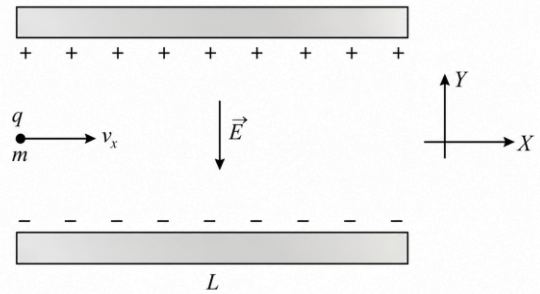


10. A particle of charge q and mass m is suspended from a point on the wall by a rigid massless rod of length $L = 3/m$ as shown. Above the point of suspension another particle is clamped which has a charge $-q$ at a distance L from point of suspension. On slight displacement from the mean position, the suspended particle is observed to execute SHM. Find the time period of SHM. (For calculations consider $Kq^2 = 2mgL^2$ and $g = \pi^2$).

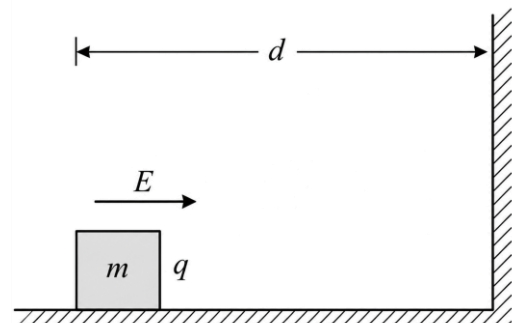


11. A uniform electric field of intensity $E = 10^6$ V/m exist in vertically downwards direction in a region. A particle of mass $m = 0.01$ kg and charge $q = 10^{-6}$ C is suspended by an inextensible thread of length $l = 1$ m. The particle is displaced slightly from its mean position and released. Calculate the time period of its oscillation. What minimum velocity should be given to the particle at rest from its equilibrium position so that it completes a full circle in vertical plane? Calculate the maximum and minimum tension in the thread in its circular motion in vertical plane.

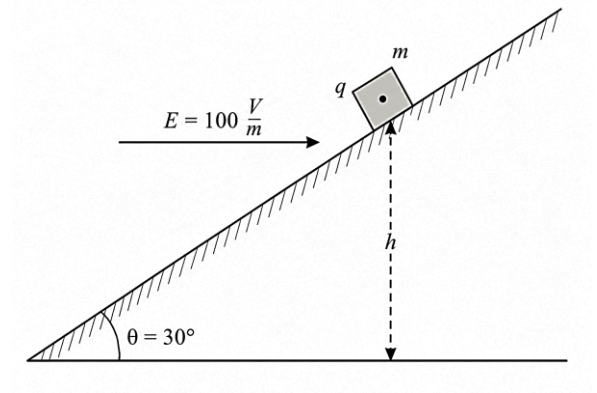
12. Figure shows an assembly of deflecting plates A and B of an ink-jet printer which causes moving ink droplets to deflect at desired displacements by continuously varying electric field between the plates. An ink drop with a mass $m = 1.3 \times 10^{-10}$ kg and a negative charge of magnitude $q = 1.5 \times 10^{-13}$ C enters the region between the plates, initially moving along the x -axis with speed $v_x = 18$ m/s. The length of plates is $L = 1.6$ cm. The plates are connected with a varying voltage and thus produce an electric field at all points between them. Assume that field $\vec{E}$ for some duration is constant and it is acting in downward direction as shown and has a magnitude of $E = 1.4 \times 10^6$ N/C, find the vertical deflection of the drop at the far edge of the plate? As the gravitational force on the drop is very small relative to the electrostatic force acting on the drop, it can be neglected for this analysis.



13. A block of mass m with a positive charge q is placed on a smooth horizontal table which terminates in a vertical wall as shown in figure. The distance of the block from the wall is d . A horizontal electric field E is switched on in the space in rightward direction. Consider elastic collisions with the wall, find the time period of resulting oscillatory motion of the block. Analyse the motion and state if it is a simple harmonic motion.



14. An inclined plane making an angle 30° with the horizontal is placed in a uniform horizontal electric field E of 100 V/m as shown in figure. A small block of mass 1 kg and charge 0.01 C is allowed to slide down from rest from a height $h = 1 \text{ m}$. If the coefficient of friction is 0.2 , find the time it will take the block to reach the bottom of incline.



Answer Key

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|---|---|
| <p>1. (a) $-\frac{2KQqx}{(R^2 + x^2)^{3/2}},$</p> <p>(b) $x = \pm x_0,$</p> <p>(c) $-\omega x_0 \sin \omega t,$</p> <p>(d) $\pi \sqrt{\frac{mR^3}{2KQq}}$</p> <p>2. $\frac{\lambda q}{4\pi\epsilon_0 R}$</p> <p>3. (2)</p> <p>4. (1, 2, 3)</p> <p>5. (1)</p> | <p>6. (3)</p> <p>7. (1, 2, 3, 4)</p> <p>8. $2\sqrt{2} \times 10^6 \text{ ms}^{-1}$</p> <p>9. $q = \frac{mg \tan \theta}{A + B \tan \theta}$</p> <p>10. (4)</p> <p>11. 0.6 s, 23.42 m/s, 6.59 N, 0</p> <p>12. $6.4 \times 10^{-4} \text{ m}$</p> <p>13. $2\sqrt{\frac{2md}{qE}}$</p> <p>14. 1.345s</p> |
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